The Shining Silverware

Time limit per test: 2 seconds
Memory limit per test: 1 gb

For thirty-six long years, the Shining Silverware has slept inside the Hall of Legends, waiting for the one winter when its light would return.

An old prophecy carved above the doors of the hall says that the Silverware can only be awakened in the year **2022**, and only by the harmony of that year.

A young wizard from the far south arrives at the hall carrying **N** ancient relics and places them in a straight line, numbered from **1** to **N** from left to right. The relic at position **i** holds a magical power a_i.

Two relics at positions **i** and **j**, where **i** < **j**, are said to **resonate** if the sum of their powers is divisible by 2022, that is:

$$(a_i + a_j) \bmod 2022 = 0$$

Every resonating pair releases a spark of light onto the Silverware.

The Council of Elders does not yet know how many relics the wizard will carry into the final ritual, so they ask **Q** questions. In each question they give an integer **k** and want to know: if only the first **k** relics (positions 1, 2, ..., **k**) are brought into the ritual, how many resonating pairs are there among them?

Help the wizard answer every question of the Council.

Input:

The first line contains a single integer **t** (1 <= **t** <= 10^4) representing the number of test cases.

Each test case consists of:

1) A line containing two integers **n** and **q** (1 <= **n**, **q** <= 2 x 10^5) representing the number of relics and the number of questions.

2) A line containing **n** integers a_1, a_2, ..., a_n (1 <= a_i <= 10^{18}) representing the powers of the relics.

3) **q** lines, each containing a single integer **k** (1 <= **k** <= **n**) describing one question.

It is guaranteed that the sum of n over all test cases does not exceed 2 x 10^5, and the sum of q over all test cases does not exceed 2 x 10^5.

Output:

For each question, print a single integer on its own line: the number of resonating pairs (**i**, **j**) with 1 <= **i** < **j** <= **k**.

Print the answers in the order the questions are given, test case after test case.

Example:

Input
3 6 4 2021 2023 2022 1011 3033 4044 6 1 5 2 4 2 1011 1011 1011 1011 4 3 1 1 2022 1
Output
3 0 2 1 6 3 0

Explanation:

Test case 1

Exactly three pairs resonate:

$$(1, 2): 2021 + 2023 = 4044 = 2 \times 2022$$

$$(3, 6): 2022 + 4044 = 6066 = 3 \times 2022$$

$$(4, 5): 1011 + 3033 = 4044 = 2 \times 2022$$

For **k** = 6, all three pairs lie among the first 6 relics, so the answer is 3.

For **k** = 1, there is only one relic, so no pair can be formed and the answer is 0.

For **k** = 5, the pairs (1, 2) and (4, 5) are counted, so the answer is 2.

For **k** = 2, only the pair (1, 2) is counted, so the answer is 1.

Test case 2

Any two relics give 1011 + 1011 = 2022, so every pair resonates. For **k** = 4 there are 6 pairs, and for **k** = 3 there are 3 pairs.

Test case 3

There is only one relic, and a relic cannot resonate with itself, so the answer is 0.

Notes:

A pair must consist of two relics at different positions, and the first must appear before the second.

Questions do not change the relics; each question is answered independently.